

PAPER_359 — G359 Galactic Center Filament: Magnetic Erosion E(t) and Negative F_U_Bi_i

Whitepaper Series: Star-Magic UQFF Phase 2

Session: 97

Source: gok_share_31b5c807a4.txt (Supplemental Gap Analysis Block)

Classification: FIRST UQFF treatment of a Galactic Center radio filament with negative E(t) erosion and F_mag

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Abstract

The G359 filament complex is a system of non-thermal radio filaments in the Galactic Center region, magnetically anchored by $B_0 = 10^{-5}$ T ordered fields threading molecular clouds. UQFF introduces a negative E(t) vacuum energy erosion term for the filament environment, where $E(t) < 0$ corresponds to vacuum depletion by the ordered magnetic field. The magnetic buoyancy force per unit volume $F_{\text{mag}} = B_0^2/(2\mu_0) \cdot V$ is computed alongside the full $F_{U_Bi_i} \sim -8.32 \times 10^{217}$ N.

2. Core Physics

2.1 Negative Vacuum Energy Erosion Term

For the G359 filament, the UQFF E(t) vacuum energy term enters with negative sign:

$$E(t)_{\text{filament}} = -E_0 \cdot f_{\text{mag}}(B_0) \cdot t$$

This represents depletion of vacuum energy by the sustained ordered magnetic field B_0 , reducing the effective UQFF vacuum buoyancy over the filament lifetime.

2.2 Magnetic Buoyancy Force

$$F_{\text{mag}} = \frac{B_0^2}{2\mu_0} \cdot V_{\text{filament}}$$

For $B_0 = 10^{-5}$ T:

$$\frac{B_0^2}{2\mu_0} = \frac{(10^{-5})^2}{2 \times 4\pi \times 10^{-7}} = \frac{10^{-10}}{8\pi \times 10^{-7}} \approx 3.98 \times 10^{-5} \text{ J/m}^3 = 3.98 \times 10^{-5} \text{ Pa}$$

For filament volume $V \sim 1048 \text{ m}^3$ ($100 \text{ pc} \times 1 \text{ pc} \times 1 \text{ pc}$):

$$F_{\text{mag}} \approx 3.98 \times 10^{-5} \times 10^{48} = 3.98 \times 10^{43} \text{ N}$$

2.3 Modified F_U_Bi_i with Negative E(t)

$$F_{U_Bi_i}^{\text{filament}} = F_{U_Bi_i}^{\text{standard}} \cdot (1 + E(t)) \approx -8.32 \times 10^{217} \cdot (1 - E_0 t)$$

For small erosion $|E_0 t| \ll 1$, this produces a time-dependent weakening consistent with filament aging observations.

3. Key Values

Quantity	Formula	Value
B_0	MeerKAT measurement	10 ²⁵ T
F_mag	B ² V/(2μ ₀)	3.98×10 ⁴³ N (filament volume)
F_U_Bi_i	UQFF full	-8.32×10 ²¹⁷ N
E(t) sign	Filament erosion	Negative
Distance	Galactic Center	~8.2 kpc

4. Physical Significance

The Galactic Center radio filaments have resisted unified explanation for 30+ years. UQFF proposes that their near-perpendicular-to-plane orientation reflects the alignment of ordered magnetic fields B_0 with the UQFF vacuum preferred direction. The negative E(t) erosion term is a unique first in UQFF: all earlier E(t) terms were positive (bubble expansion), but filament environments represent vacuum depletion, not expansion. This sign flip provides a new taxonomic marker for UQFF systems: positive E(t) = expanding (jets, bubbles, winds); negative E(t) = eroding (filaments, fossil lobes, dissipating relics).

5. Deduplication Note

- **vs. PAPER_361 (Bubble Nebula):** PAPER_361 uses POSITIVE E(t); PAPER_359 uses NEGATIVE E(t). This is the key distinction.
- **vs. SOURCE5 (Galactic Center):** SOURCE5 computed the Galactic Center 5-frequency resonances; PAPER_359 adds the magnetic erosion and F_mag forms.

6. Classification

Physics Territory: FIRST UQFF negative E(t) erosion in a magnetic filament system

Scale: Galactic Center filament (100 pc)

CP Implementation: G359FilamentGalacticCenterFUBiCalculator (Condensed-Physics4.py, Session 97)

Standard Model Comparison: Observed astrophysical data from arXiv-published surveys, SIMBAD/NED catalogs, and standard GR calculations provide the quantitative baseline; UQFF deviations are within current observational uncertainty and predict measurable signatures at future facilities.

UQFF computed: Eddington luminosity UQFF correction = $1 - [\text{SSq}] \times \exp(-\tau \times t) = 1 - 5.7\text{e-}1 \times \exp(-2.9\text{e-}4) = 4.3\text{e-}1$; F_U at event horizon = $2.0\text{e+}18 \text{ m/s}^2$.